www.PapaCambridge.com

O

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

*8413371862*

**ENVIRONMENTAL MANAGEMENT** **5014/01**

Paper 1 **May/June 2007**

**2 hours 15 minutes**

Candidates answer on the Question Paper.

Additional Materials: Ruler

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
All questions in Section A carry 10 marks.
Both questions in Section B carry 40 marks.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| Total | |

This document consists of **23** printed pages and **1** blank page.


© UCLES 2007

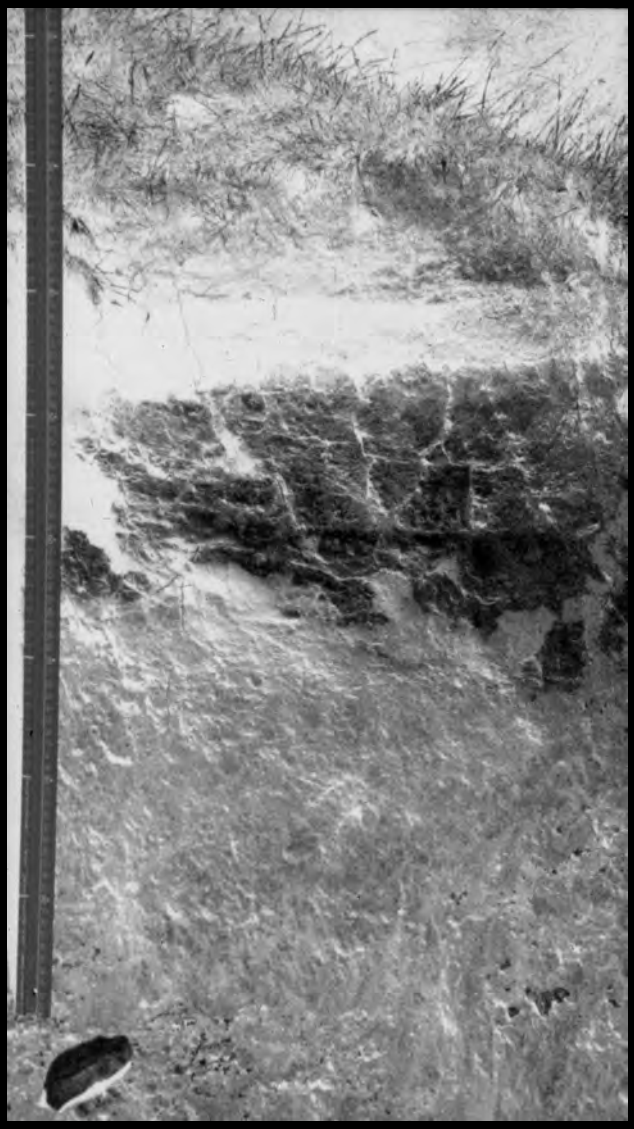
www.PapaCambridge.com

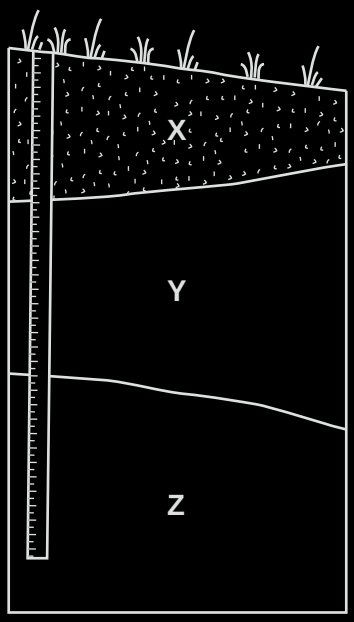


For Examiner's Use

## Section A

**1** **(a)** The photograph shows a soil formed on a rock made of sand and the diagram is a profile of the layers of the soil from the surface vertically downwards.






scale x 0.5

**(i)** Which layer, **X** or **Y**, has the most organic matter? State the evidence for this.

layer .....

evidence .....[2]

**(ii)** **Z** is parent material which has broken down. What is the evidence for this?

.....[1]

www.PapaCambridge.com

For Examiner's Use

**(b)** The soil in the photograph contains a high percentage of sand and has a coarse
It has a pH of 4. Explain:

**(i)** why the soil has a coarse texture,

.....[1]

**(ii)** what a pH of 4 indicates,

.....

.....[2]

**(iii)** how the texture of a soil and the size of its pore spaces are linked.

.....

.....[1]

**(c)** **(i)** State one advantage and one disadvantage, for crop growth, of a soil with a coarse texture.

advantage .....

disadvantage .....[2]

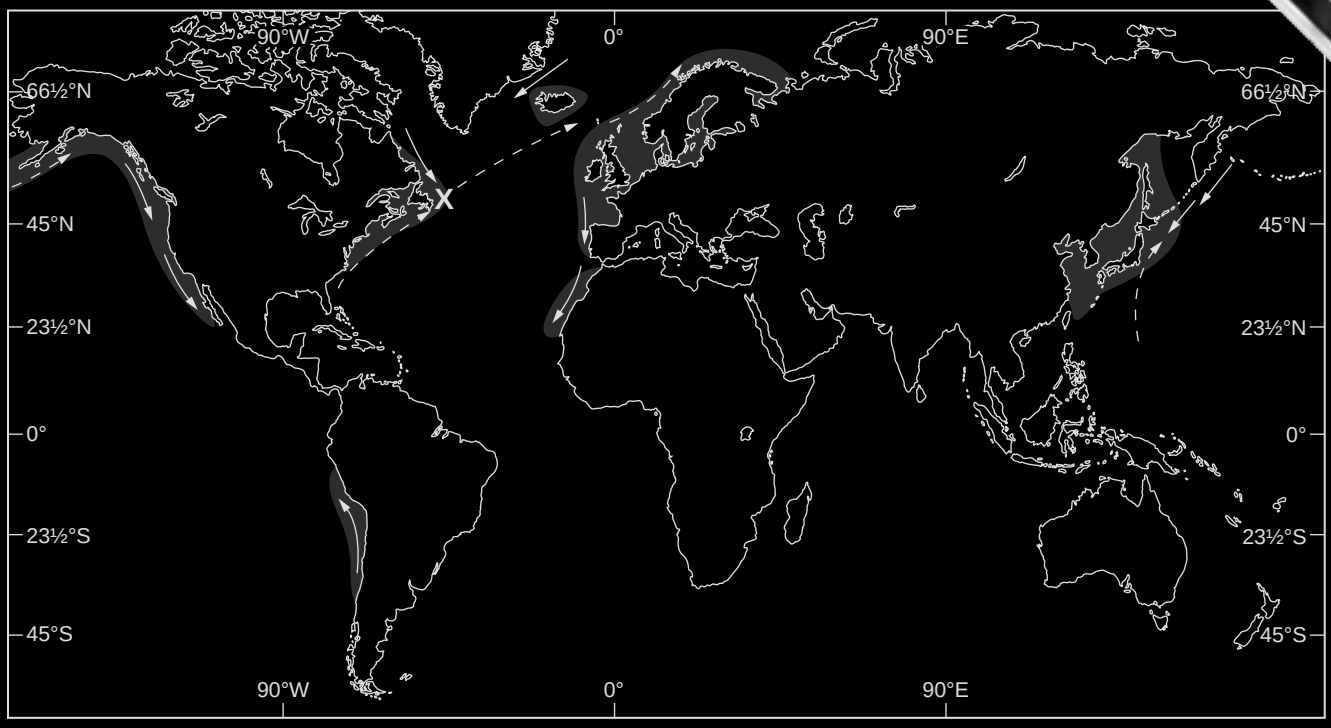
**(ii)** State **one** disadvantage of a pH of 4.

.....[1]

www.PapaCambridge.com

For Examiner's Use

**2** **(a)** The map shows some of the world's main marine fishing grounds.




**Key**

fishing grounds

cold ocean current

warm ocean current

What **three** features of location do they all have in common?

.....

.....

.....[3]

**(b)** Describe the conditions which favour large marine fish populations.

.....

.....

.....

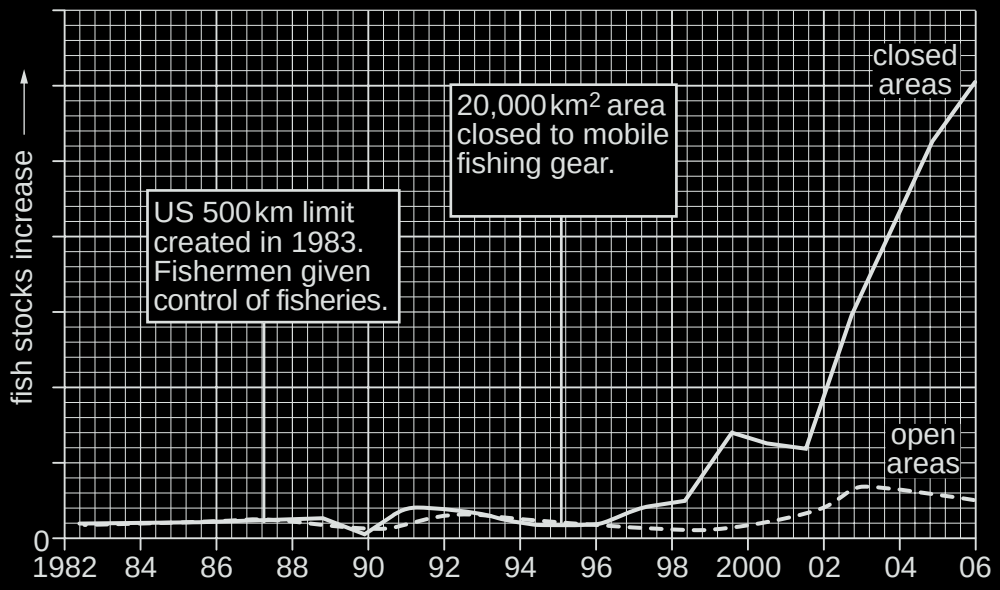
.....

.....[3]

www.PapaCambridge.com

For Examiner's Use

**(c)** When fish stocks collapsed, in area **X**, measures were taken to encourage the re of the industry. The graph below shows the timing of these measures and their ef on the fish population.




Describe, and suggest reasons for, the different effects of each of the two measures.

.....

.....

.....

.....

.....

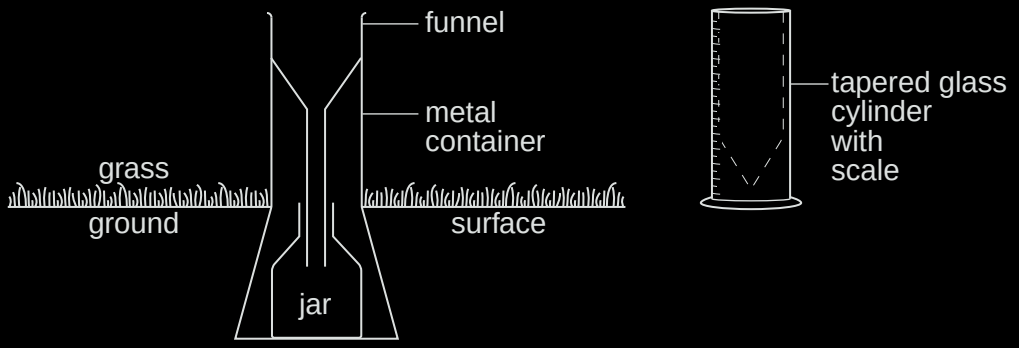
.....

.....[4]

www.PapaCambridge.com

For Examiner's Use

**3** **(a)** **(i)** The diagram shows the instruments used to measure precipitation at a w station.




How would you take accurate measurements of the precipitation using these instruments?

.....

.....

.....

.....[2]

**(ii)** Explain why this instrument is sited as shown in the diagram to give accurate readings of the precipitation.

.....

.....

.....

.....[2]

**(iii)** Why is it sometimes more difficult to measure precipitation accurately in places with a tundra climate?

.....[1]

www.PapaCambridge.com

For Examiner's Use

**(b)** Describe the amount, frequency and intensity of precipitation in a desert climate.

.....

.....

.....[3]

**(c)** Explain why trickle drip irrigation is the least harmful type of irrigation to use in a desert climate.

.....

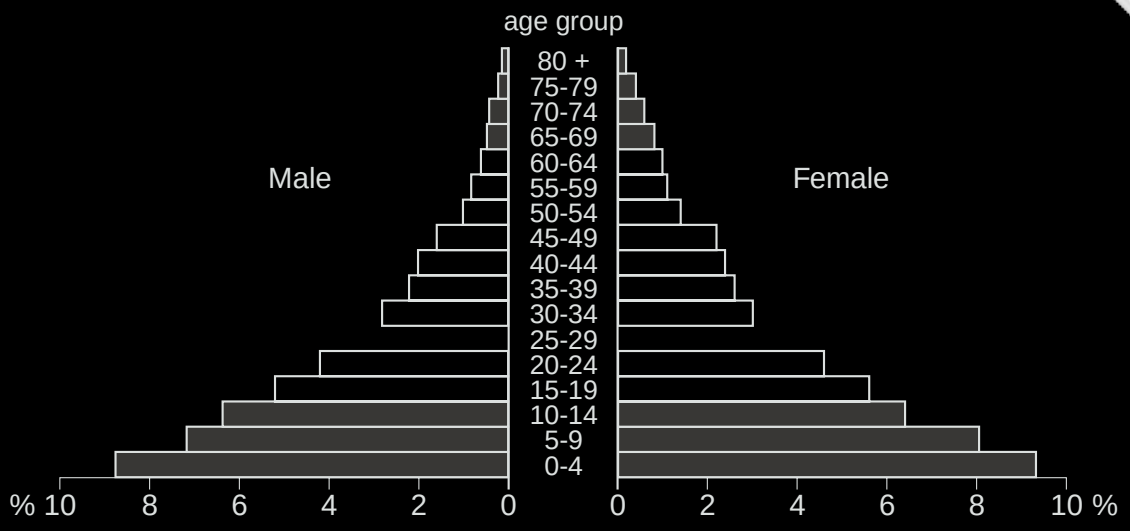
.....

.....[2]

www.PapaCambridge.com

For Examiner's Use

**4** **(a)** Look at the population pyramid of Mexico.




**(i)** What percentage of the country's population is below five years old?

..... % [1]

**(ii)** The table shows data missing from the population pyramid.

| age group | males | females |
|---|---|---|
| 25–29 | 3.2% | 3.8% |

Complete the population pyramid for the 25–29 age group. [1]

**(iii)** Circle the words which describe how the country's population is changing now. Choose from:

decreasing slowly decreasing rapidly not changing

increasing slowly increasing rapidly. [1]

**(b)** Describe and explain the problems for a country with this type of population structure.

.....

.....

.....

.....

.....

.....

.....[4]

www.PapaCambridge.com

For Examiner's Use

**(c)** Why is it difficult for governments to introduce policies to change this pop structure?

.....

.....

.....

.....

.....[3]

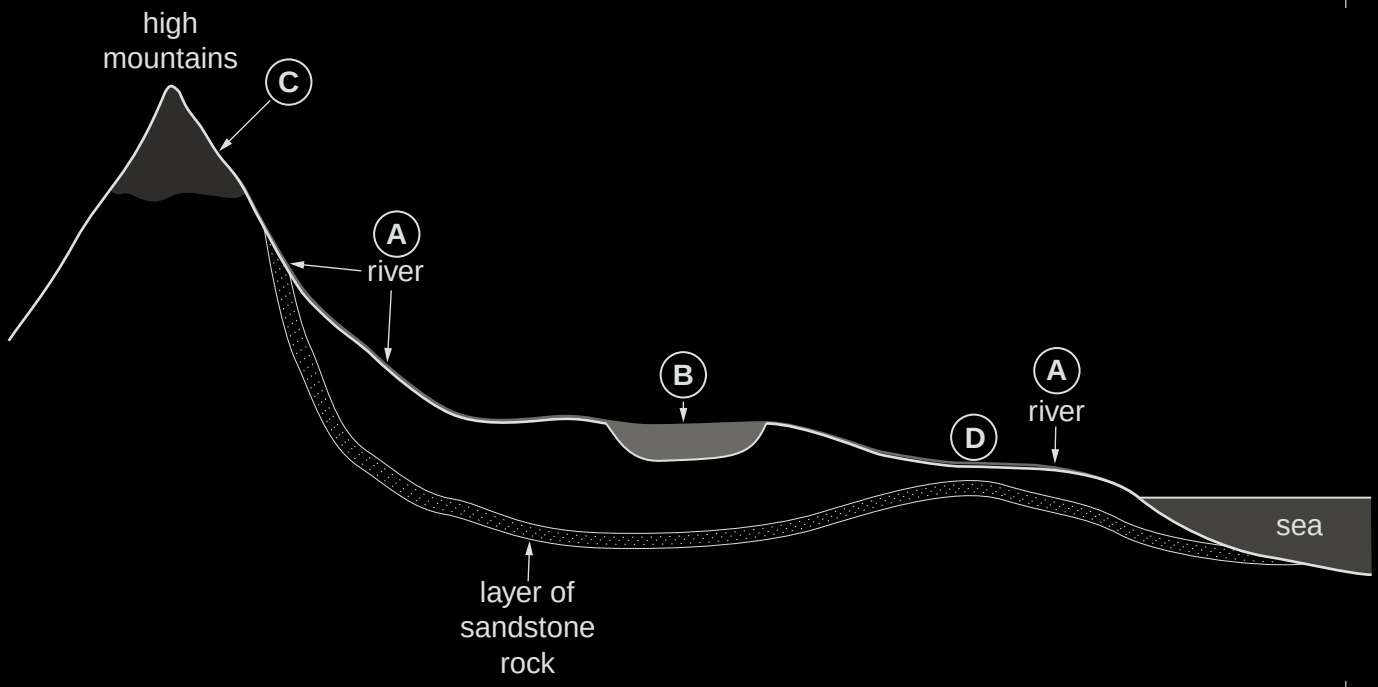
www.PapaCambridge.com

For Examiner's Use

## Section B

**5** **(a)** Look at the cross-section.

**Section from the mountains to the sea**




**(i)** Name the **two** stores of fresh water labelled **B** and **C** on the section.

**B** .....

**C** .....[2]

**(ii)** Why might it be possible for people to obtain fresh water, at point **D**, other than from the river?

.....

.....

.....

**(iii)** How could people obtain this water?

.....

.....

.....

.....
[3]

www.PapaCambridge.com

For Examiner's Use

**(iv)** Choose **one** of the sources from **A**–**D** which is likely to have water that will ... and clean for people to use. Explain your choice.

Letter .....

Explanation .....

.....

.....[2]

**(v)** Which one of the four sources is **least** likely to give clean water? Explain your choice.

Letter .....

Explanation .....

.....

.....

.....

.....[3]

**(b)** Dams are often built to hold back reservoirs for water supply.

**(i)** Give the name and location of a dam.

.....

**(ii)** Explain why the dam was built and what advantages it has brought to the local people.

.....

.....

.....

.....

.....

.....

.....

.....

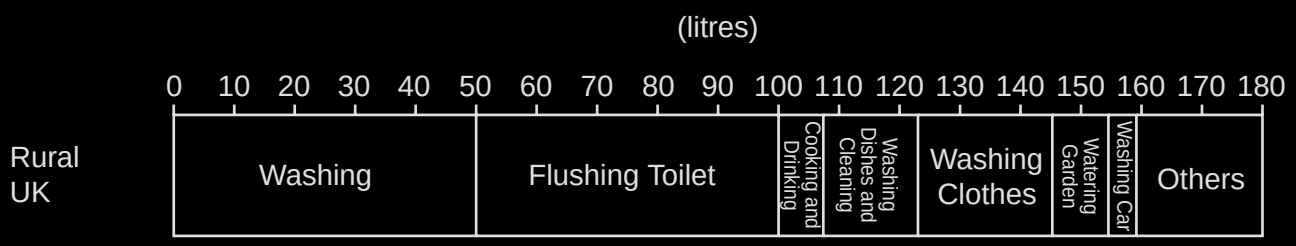
.....[5]

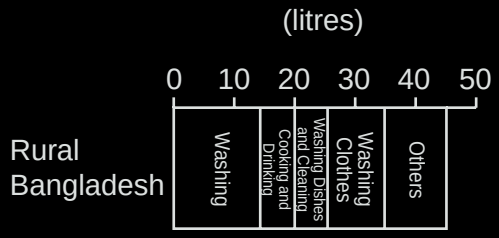
www.PapaCambridge.com

For Examiner's Use

**(c)** The divided bar graphs show average amounts of water used, by a family each rural areas of the UK and Bangladesh. They also show how the water is used.

**Average daily water use and uses by a family in rural areas of the UK and Bangladesh**




**(i)** How many more litres of water are used by a family in the UK than in Bangladesh?

.....[1]

**(ii)** How many times greater is the amount used by a family in the UK?

.....[1]

**(iii)** For which one of the uses named in the graphs is the daily consumption of water almost the same in the UK and Bangladesh?

.....[1]

**(iv)** Suggest a reason for this.

.....

.....[1]

**(v)** The UK is a developed country. Bangladesh is a developing country.
Explain why there are differences in use of water between countries with different levels of economic development.

.....

.....

.....

.....

.....

.....

.....

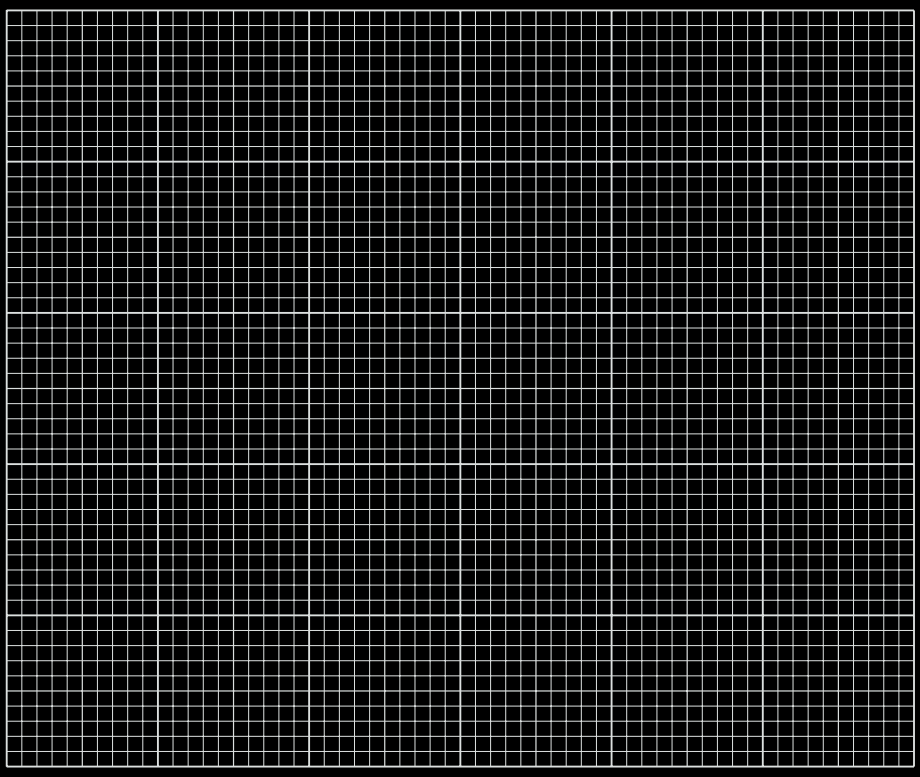
www.PapaCambridge.com

For Examiner's Use

**(d)** There are large differences in access to safe water supplies between urban a areas in many developing countries in Africa. Two examples are shown in the table.

| % of people with access to safe water supplies | | |
|---|---|---|
| Country | Urban areas | Rural areas |
| Kenya | 70 | 50 |
| Nigeria | 85 | 40 |

**(i)** On the graph paper below, draw a bar graph to show these percentages. Complete your graph with a key.



[4]

**(ii)** Give reasons why there are differences in access to safe water supplies between urban and rural areas in many developing countries.

.....

.....

.....

.....

.....

.....[3]

www.PapaCambridge.com

For Examiner's Use

**(e)** Look at the cartoon about a rural area in Africa.

**Walking four hours to fetch water**




**(i)** What does it suggest about the views of some people in the area to change and development?

.....

.....

.....

.....[2]

www.PapaCambridge.com

For Examiner's Use

**(ii)** Explain how sinking a new well can reduce disease, increase economic and improve the quality of life for everyone in rural areas in developing coun especially for the women and children.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....[5]

**(iii)** Why is outside help from charities such as Water Aid often needed?

.....

.....

.....

.....

.....

.....[3]

[Total: 40]

www.PapaCambridge.com

For Examiner's Use

**6** **(a)** Look at the two plate boundaries shown in Diagrams **A** and **B**.

**Diagram A**

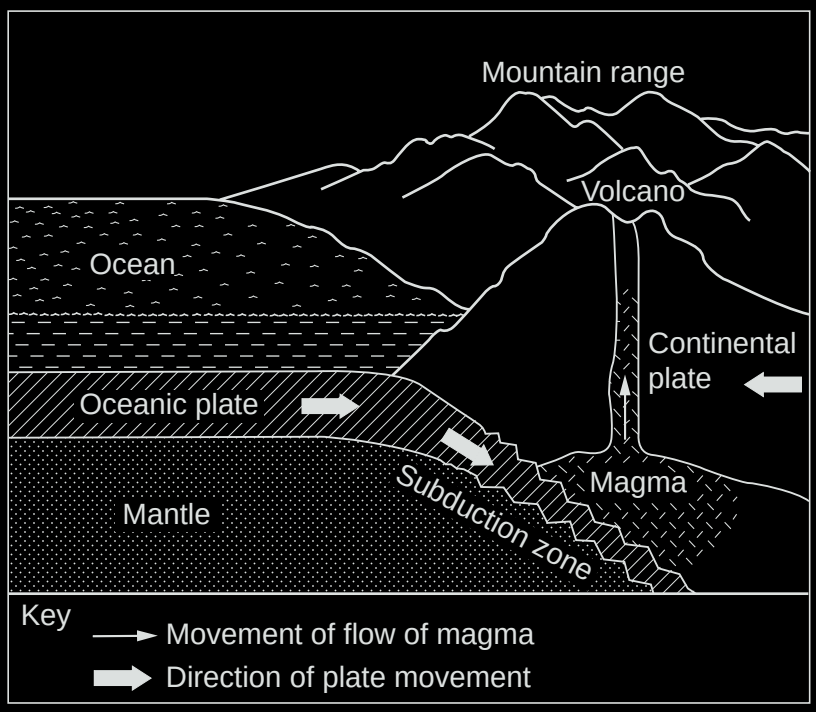
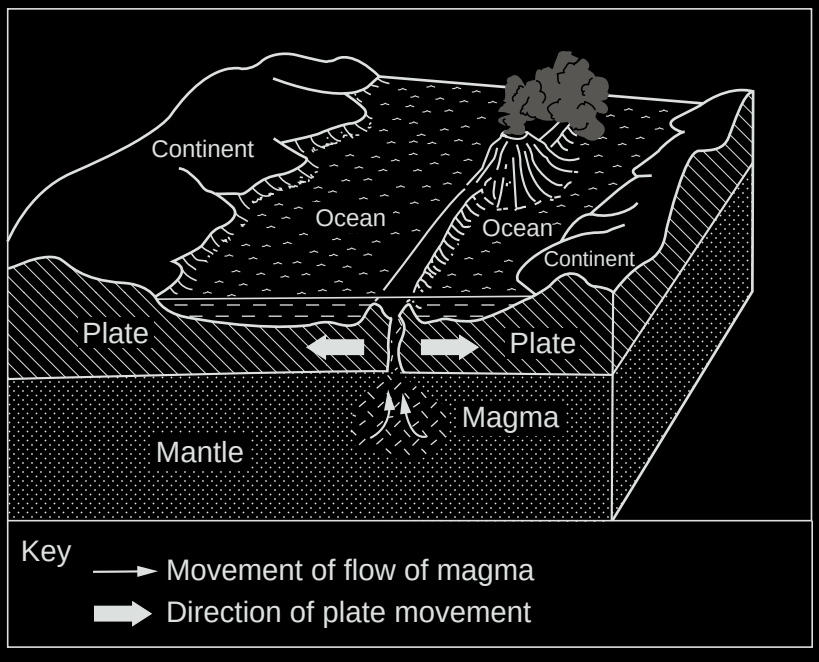



**Diagram B**




**(i)** Describe what is happening to the two plates in **A** and the two plates in **B**.

**A** .....

.....

**B** .....

[2]

www.PapaCambridge.com

For Examiner's Use

**(ii)** The source of the magma is different in the two diagrams. Where has the magma come from in **A** and **B**?

**A** .....

.....

**B** .....

.....[3]

**(iii)** Why are volcanoes formed at plate boundaries?

.....

.....

.....

.....[2]

**(iv)** State **one** difference between volcanoes along these two types of plate boundary.

.....

.....

.....[2]

**(v)** Some volcanic eruptions result in great loss of life; in others, no one is killed. Give reasons for the large difference in numbers of people killed.

.....

.....

.....

.....

.....

.....

.....

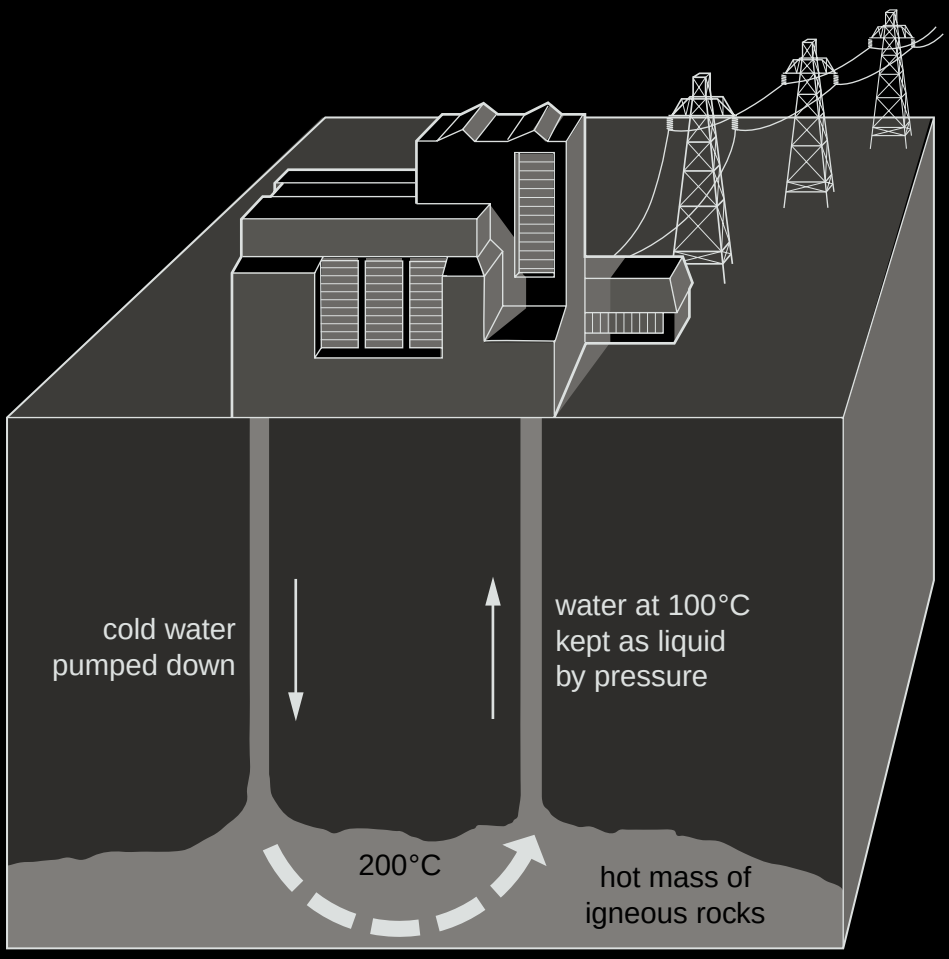
.....[4]

www.PapaCambridge.com

For Examiner's Use

**(b)** In some countries electricity is made from geothermal power. Look at the diagram of a geothermal power station.

**Geothermal power**




**(i)** How is the electricity produced?

.....

.....

.....

.....

.....

**(ii)** Why are areas of active volcanic activity needed for its production?

.....

.....

.....

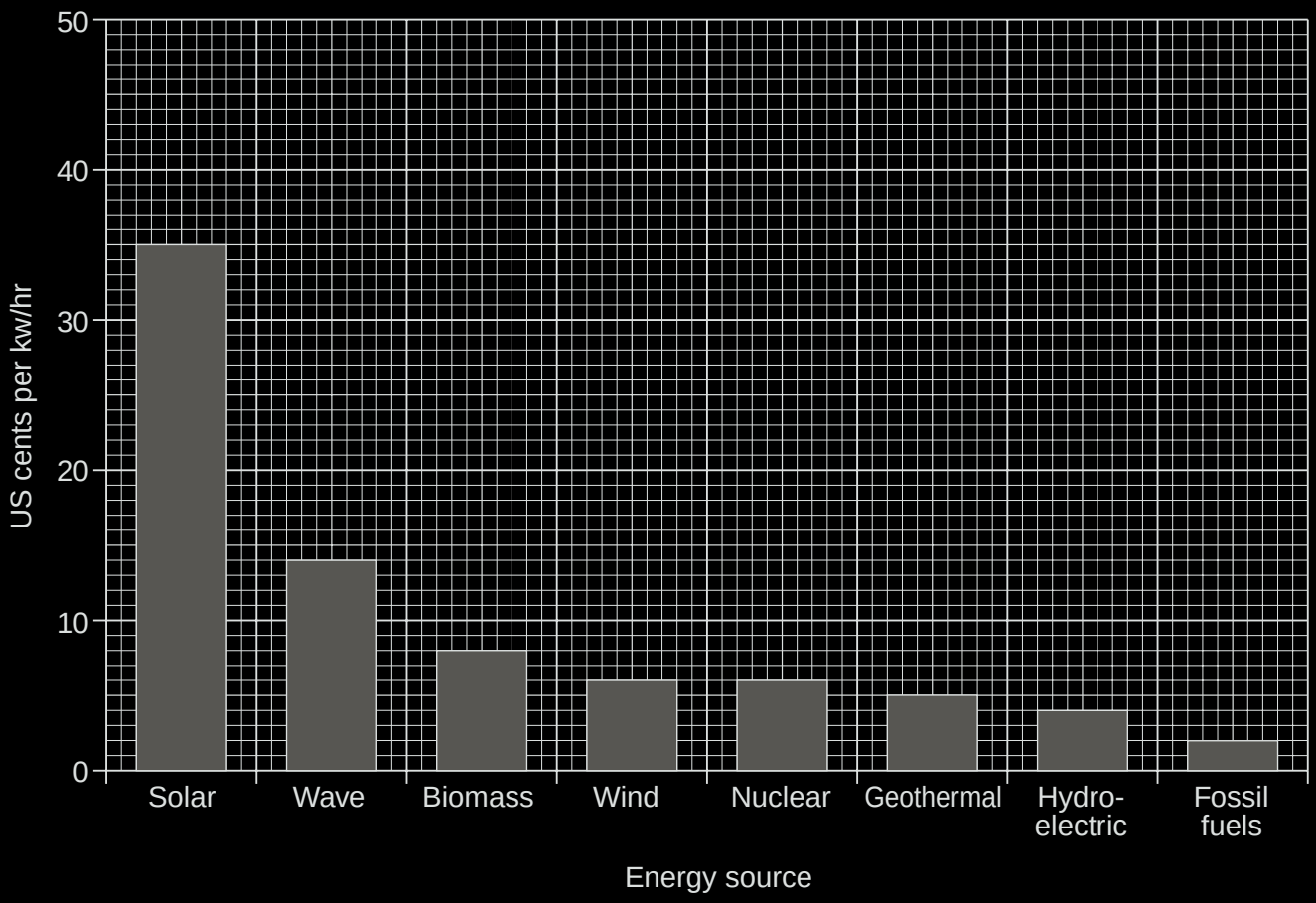
[4]

www.PapaCambridge.com

For Examiner's Use

**(c)** The graph shows average costs of producing electricity from different energy sou
2003.

**Costs of producing electricity (2003)**




**(i)** Describe what the graph shows about the cost of producing geothermal electricity compared with electricity from other energy sources.

.....

.....

.....

.....

.....[3]

**(ii)** Explain how likely it is that geothermal power will be used more in the future as an alternative to fossil fuels.

.....

.....

.....

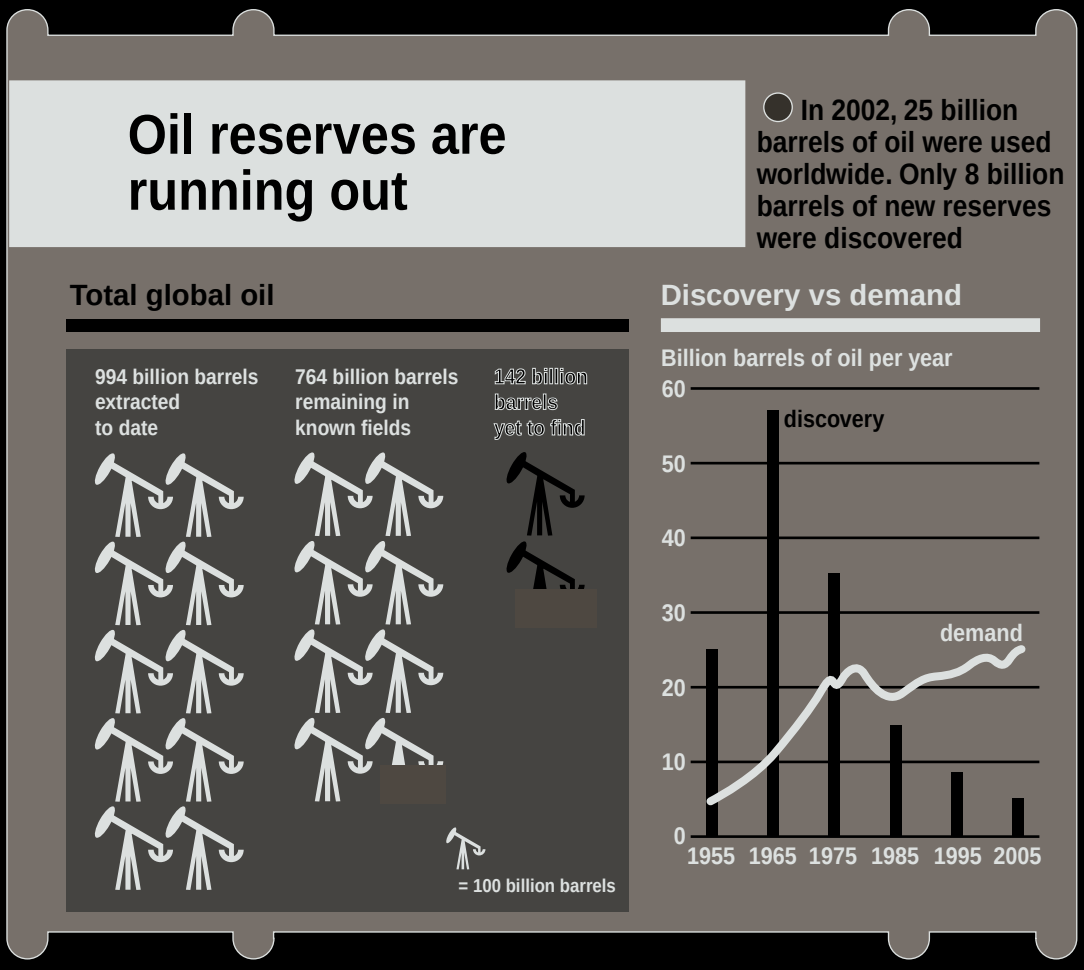
.....[2]

www.PapaCambridge.com

For Examiner's Use

**(d)** Look at the information below about world supply and demand for oil.

**Oil reserves are running out**




**(i)** Describe what happened to the demand for oil between 1955 and 2005.

.....

.....

.....

.....

.....[3]

www.PapaCambridge.com

For Examiner's Use

**(ii)** Quote values and information which support the following statements.

**1** Oil reserves are running out.

.....

.....

.....

.....

**2** The present use of oil is not sustainable.

.....

.....

.....

.....[4]

**(iii)** What is likely to happen to the demand for oil after 2005? Explain your answer.

.....

.....

.....

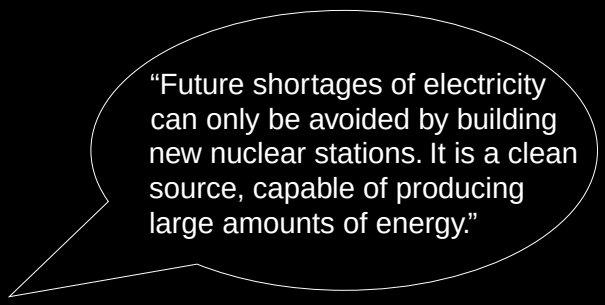
.....

.....[3]

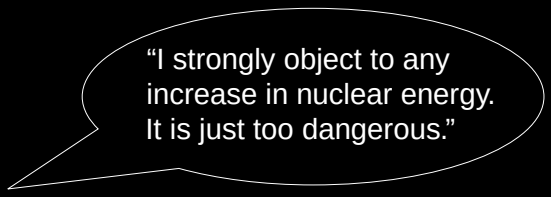
www.PapaCambridge.com

For Examiner's Use

**(e)** **A** One person's view of nuclear energy




**B** Another person's view of nuclear energy.




**(i)** Explain why some people hold the view about nuclear energy stated in **A**.

.....

.....

.....

.....

.....

.....[3]

**(ii)** State the different arguments which supporters of view **B** could use.

.....

.....

.....

.....

.....

.....[3]

www.PapaCambridge.com

For Examiner's Use

**(iii)** What is your view on nuclear power? How strong are the different argume... forward in part **(ii)**?

.....

.....

.....

.....

.....[2]

[Total: 40]

**BLANK PAGE**

www.PapaCambridge.com

*Copyright Acknowledgements:*

Question 1(a) M. Fretwell © UCLES.
Question 2(d) © The Association for the Study of Peak Oil & Gas.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

University of Cambridge International Examinations is part of the University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of